

Test 2

EEC 3150

For each exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

Test<test number>_Ex<exercise number>_<FirstInitial><LastName>.py

Example: Test01_Ex01_GWest.py

Exercise 1 (30 points)

For this exercise you will need to use `getcwd()` and `listdir()` from the `os` module.

- a. Write a function `DirLineCount(path)` which will return a dictionary whose keys are the filenames of all files in the `path` directory and whose values are the number of lines in each respective file.
- b. Demonstrate your function by calling it on the current working directory. Print the results.

Exercise 2 (30 points)

In this exercise, you will read and extract content from an HTML file. Please view the file before attempting to code the problem. To view an HTML file, you must open it with a text editor (opening it by simply double-clicking will default to using the browser).

- a. Write a function `GetImages(fname)` which will read the HTML file `fname` and return a list of image filenames that the HTML page displays.
- b. Test your function on the `image.html` file. Print the list of image filenames.

TIP: In HTML, images are placed onto the page using `<img>` tags. The file that they display are given by the `src` attribute (short for “source”). For example, in the tag

```
<img id='img1' src='image1.jpg'>
```

the file `image1.jpg` is what will be displayed on the screen.

Exercise 3 (40 points)

In this exercise, you will create a module `data.py` in which you define a class `Data` and then import it into another module where you will use it.

- a. Create a Python module named `data.py`. In this module, define the class `Data` (notice the difference in capitalization b/t the two). This class should...
 - have a constructor which accepts a list of numeric data (`int`, `float`) and stores it as an attribute called `data`
 - have a `__str__` magic method which will simply return the `data` attribute, but cast to a `str`
 - have methods which compute the mean and standard deviation
- b. Import the `data.py` module into another module (named according to our standard format for HW/test exercises). Demonstrate it by creating a `Data` object and printing its `data` attribute, mean, and standard deviation.

For reference, the equation for standard deviation is given by

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}$$

where N is the number of data points and $\bar{x}$ is the mean.